

Committee Review Package

Nightly: PASS [PASS] latest release chain

Nightly Smoke: PASS [PASS] profile=balanced_practice

Smoke Pass Rate: 100.00% [PASS] count=10

Smoke Trial Feasible: 100.00% [PASS] avg_trial_runtime_s=0.0563

Smoke Strict: candidate_for_strict_enable [PASS] history count and pass/feasible rates satisfy strict-ready threshold

CI Gate: PASS [PASS] static + contract gate | MIDAS section-library: ok | 182/183 used | 183 templates | source=midas_parser_derived | implement

PBD Drift Max: 9.2000% [PASS] 7-GM envelope

PBD Binary Cases: 7 [PASS] npz_external+inline_summary

Speedup: 18,818x [PASS] vs 336h estimate

Wind Duration: 10.0h [PASS] across-wind benchmark

SSI Ratio Span: 0.2823 [PASS] nonlinear boundary

Damper Corr Min: 0.9625 [PASS] NHERI damped frame

Diff. Shortening: 38.331 mm [PASS] construction sequence

Flex Ampl. Max: 1.200 [PASS] shell-beam mix

Replay Runs: 3 [PASS] version lock

Signed Registry: ed25519 [PASS] release traceability | project_pkg=3121538B | approvals=2

KDS PASS: 496 [PASS] code check rows

NDTHA Residual: 1.914% [PASS] hard-threshold gate

SAC Holdout: 3 [PASS] global benchmark

NHERI Holdout: 3 [PASS] sensor waveform

OpenSees Holdout: 3 [PASS] topology authority

OpenSees Canonical Breadth: 7 [PASS] families=6_parser_ready=3

PBD / 7-GM Drift Envelope: max=9.2000%, p95=9.2000% [PASS]

criterion=drift envelope within IO/LS/CP band | evidence=pbdr_review_drift_envelope

PBD / Energy Balance: 8.427672e-05 [PASS]

criterion=relative error $\leq 1e-2$ | evidence=killshot_metrics

PBD / Binary artifact trace: storage=npz_external+inline_summary, cases=7 [PASS]

criterion=npz metrics artifact present and case count matches selected set | evidence=pbdr_review_package_report

Optimization / Long-Budget Feasible Repair: feasible=True, max_dcr=0.9332 [PASS]

criterion=solver_feasible_final == true and final max DCR ≤ 1.0 | evidence=design_optimization_solver_loop_long_report

Optimization / Cost Reduction Evidence: changed=20, accepted=23, cost_delta=411.159 [PASS]

criterion=feasible input, blocked=false, cost reduction nonnegative | evidence=design_optimization_cost_reduction_report

Optimization / Blocked Action Explain: blocked=47, illegal=0, no_gain=8 [PASS]

criterion=rejected cost-down actions are exported with machine-readable reasons | evidence=design_optimization_cost_reduction_blocked_actions

NDTHA / Residual Hard Gate: top=0.688 m, drift=1.914%, fallback_rate=0.000 [PASS]

criterion=finiteness residuals, traceable source, hard thresholds pass | evidence=ndtha_residual_gate_report

NDTHA / Residual Trace: cases=3, changed=3, pre_top_max=3.308 m, pre_drift_max=9.200% [PASS]

criterion=pre-settle and post-settle residuals remain auditable per case | evidence=nonlinear_ndtha_stress_report

Wind / Across-wind long series: duration=10.0h, chunks=72, post_drift=0.000%, pre_drift=0.001% [PASS]

criterion=10h series, converged, no collapse | evidence=wind_time_history_gate_report

SSI / Nonlinear boundary response: ratio span=0.2823, fixed_post_drift=0.033%, ssi_post_drift=0.004% [PASS]

criterion=finiteness transfer and shear delta pass | evidence=ssi_boundary_gate_report

Damper / Damped frame agreement: corr_min=0.9625, phase_max=30.000 ms [PASS]

criterion=correlation/phase/residual thresholds pass | evidence=damper_validation_gate_report

Construction / Construction preload: diff=38.331 mm, stress=23.241 MPa [PASS]

criterion=creep/shrinkage + differential shortening captured | evidence=construction_sequence_gate_report

Diaphragm / Flexible diaphragm: amp=1.200, drift=3.087% [PASS]

criterion=shell-beam mix and slab shear checks pass | evidence=flexible_diaphragm_gate_report

Reproducibility / Version lock replay: runs=3, seed=23 [PASS]

criterion=exact replay match and manifest written | evidence=reproducibility_version_lock_report

Reproducibility / Signed release registry: artifacts=12, project_pkg=3121538B, approvals=2, pubkey=implementation/phase1/release/signing/release_r

criterion=signature verified and registry hashes present | evidence=release_registry

Code Check / KDS compliance: rows=511, pass_cards=8 [PASS]

criterion=all compliance rows pass | evidence=kds_compliance_summary

Authority / SAC holdout: cases=3, valid=3 [PASS]

criterion=drift/base shear/member force thresholds pass | evidence=global_authority_gate_report

Authority / NHERI holdout: cases=3, valid=3 [PASS]

criterion=waveform correlation/phase/residual thresholds pass | evidence=global_authority_gate_report

Authority / OpenSees holdout: cases=3, pass=3 [PASS]

criterion=real topology + shell/beam mix topology contracts pass | evidence=global_authority_gate_report

Authority / OpenSees canonical breadth: families=6, cases=7, parser_ready=3 [PASS]

criterion=committed real-source OpenSees canonical assets are surfaced beyond holdout-only coverage | evidence=opensees_canonical_breadth_report

Authority / Measured benchmark breadth: baseline=2/51, delta=6/7, measured=21/294 [PASS]

criterion=commercial measured baseline plus committed OpenSees breadth are tracked in one surfaced gate | evidence=measured_benchmark_breadth_report

PEER / Blind compare lane: cases=10, measured_response_ready=True, channels=11/11, lane=PEER Blind Prediction Compare Family [PASS]

criterion=compare lane is staged and measured-response readiness is tracked | evidence=peer_blind_prediction_compare_report

PEER / Measured-response landing: state=recorded, matched=1, csv=0, accel=0, drift=0, sensors=0 [PASS]

criterion=official measured-response bundle is staged under the landing root | evidence=edefense_peer_blind_prediction_seed_01.measured_response_landing_manifest

Authority / Binary artifact trace: storage=npz_external+inline_summary, cases=9, steps=9 [PASS]

criterion=npz metrics artifact present and case/step counts recorded | evidence=global_authority_gate_report

Authority Benchmarks

OpenSees / SCBF16B: nodes=420 | shell_beam_mix=False [PASS]

source=implementation/phase1/open_data/megastructure/opensees/SCBF16B.tcl | provenance=opensees_text

OpenSees / SCBF16B_shell_beam_mix: nodes=426 | shell_beam_mix=True [PASS]

source=implementation/phase1/open_data/megastructure/opensees/SCBF16B_shell_beam_mix.tcl | provenance=opensees_text

OpenSees / Luxinzheng_Megatall_Model1: nodes=1252 | shell_beam_mix=False [PASS]

source=implementation/phase1/open_data/irregular/collected/artifacts/luxinzheng_megatall_tcl_model1_local/opensees.tcl | provenance=opensees_text

SAC / SAC20_LA_holdout_01: drift=3.200% | MAC=0.9710 [PASS]

source=https://femap58.atcouncil.org/index.php?option=com_content&view=article&id=5:sac-steel-project&catid=2:uncategorised | provenance=309234fd42a5

SAC / SAC20_SEA_holdout_02: drift=3.600% | MAC=0.9680 [PASS]

source=https://femap58.atcouncil.org/index.php?option=com_content&view=article&id=5:sac-steel-project&catid=2:uncategorised | provenance=ceb64448b2a0

SAC / SAC20_BOS_holdout_03: drift=2.900% | MAC=0.9740 [PASS]

source=https://femap58.atcouncil.org/index.php?option=com_content&view=article&id=5:sac-steel-project&catid=2:uncategorised | provenance=309234fd42a5

NHERI / NHERI_UCSD_like_holdout_01: corr=1.0000 | phase=0.000 ms [PASS]

source=https://www.designsafe-ci.org/data/browser/public/designsafe.storage.published/ | provenance=0342dbfd730d

NHERI / NHERI_UCSD_like_holdout_02: corr=1.0000 | phase=0.000 ms [PASS]

source=https://www.designsafe-ci.org/data/browser/public/designsafe.storage.published/ | provenance=0342dbfd730d

NHERI / NHERI_UCSD_like_holdout_03: corr=1.0000 | phase=0.000 ms [PASS]

source=https://www.designsafe-ci.org/data/browser/public/designsafe.storage.published/ | provenance=0342dbfd730d

Appendix: Design Optimization Entrypoint Details

Ablation: reports 1/1, pass=1, fail=0

Dataset: reports 1/1, pass=1, fail=0

Profile: reports 1/1, pass=1, fail=0

Stage A: reports 2/2, pass=2, fail=0

Stage A/B/C: reports 1/1, pass=1, fail=0

Stage B: reports 1/1, pass=1, fail=0

Ablation (1 rows, pass=1, fail=0, reasons=PASS:1)

ablation: pass=True reason=PASS

report=/home/betelgeuze/건축구조분석/implementation/phase1/release/design_optimization/design_optimization_ablation_report.json

Dataset (1 rows, pass=1, fail=0, reasons=PASS:1)

dataset: pass=True reason=PASS

report=/home/betelgeuze/건축구조분석/implementation/phase1/release/design_optimization/design_optimization_dataset_report.json

Profile (1 rows, pass=1, fail=0, reasons=PASS:1)

objective_profile: pass=True reason=PASS

report=/home/betelgeuze/건축구조분석/implementation/phase1/release/design_optimization/design_objective_profile_report.json

Stage A (2 rows, pass=2, fail=0, reasons=PASS:2)

solver_loop: pass=True reason=PASS

report=/home/betelgeuze/건축구조분석/implementation/phase1/release/design_optimization/design_optimization_solver_loop_report.json

solver_loop_long: pass=True reason=PASS

report=/home/betelgeuze/건축구조분석/implementation/phase1/release/design_optimization/design_optimization_solver_loop_long_report.json

Stage A/B/C (1 rows, pass=1, fail=0, reasons=PASS:1)

budgeted: pass=True reason=PASS

report=/home/betelgeuze/건축구조분석/implementation/phase1/release/design_optimization/design_optimization_budgeted_report.json

Stage B (1 rows, pass=1, fail=0, reasons=PASS:1)

cost_reduction: pass=True reason=PASS

report=/home/betelgeuze/건축구조분석/implementation/phase1/release/design_optimization/design_optimization_cost_reduction_report.json

Appendix: MIDAS KDS Row Provenance Export

MIDAS KDS row provenance export: PASS | combos=6 | rows=144 | members=12 | clauses=6 | exact_rows=144 | clause_filters=6 | member_filters=12 | reverse_jump

json=implementation/phase1/release/kds_compliance/midas_kds_row_provenance_table.json | csv=implementation/phase1/release/kds_compliance/midas_kds_row_provenance_table

gLCB1 | member=MF-012 | clause=KDS-MOMENT-Y-001 | baseline=27441

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TST-003 | case=C-TST-003 | baseline=27441 | member_types=column

gLCB1 | member=MF-005 | clause=KDS-MOMENT-Y-001 | baseline=27441

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TRN-005 | case=C-TRN-005 | baseline=27441 | member_types=column

gLCB1 | member=MF-012 | clause=KDS-INT-FRAME-001 | baseline=27441

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TST-003 | case=C-TST-003 | baseline=27441 | member_types=column

gLCB1 | member=MF-010 | clause=KDS-MOMENT-Y-001 | baseline=27441

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TST-001 | case=C-TST-001 | baseline=27441 | member_types=column

gLCB1 | member=MF-012 | clause=KDS-SHEAR-Y-001 | baseline=27441

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TST-003 | case=C-TST-003 | baseline=27441 | member_types=column

gLCB1 | member=MF-005 | clause=KDS-INT-FRAME-001 | baseline=27441

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TRN-005 | case=C-TRN-005 | baseline=27441 | member_types=column

gLCB1 | member=MF-007 | clause=KDS-MOMENT-Y-001 | baseline=27425

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TRN-007 | case=C-TRN-007 | baseline=27425 | member_types=wall

gLCB1 | member=MF-012 | clause=KDS-AXIAL-001 | baseline=27441

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TST-003 | case=C-TST-003 | baseline=27441 | member_types=column

NDTHA Residual Case Trace

C-TST-001 [test/seismic]: top 3.3080->0.6881 m | drift 9.2000->1.9136%

delta_top=2.6199 m | delta_drift=7.2864% | source=solver_raw | settle_steps=48

C-TST-002 [test/seismic]: top 2.7677->0.5757 m | drift 9.2000->1.9136%

delta_top=2.1920 m | delta_drift=7.2864% | source=solver_raw | settle_steps=48

C-TST-003 [test/combined]: top 3.2918->0.6847 m | drift 9.2000->1.9136%

delta_top=2.6071 m | delta_drift=7.2864% | source=solver_raw | settle_steps=48

Blocked Cost-Down Actions

blocked=47
illegal_by_mask=0
no_cost_gain=8
constructability_hard_gate=3
no_cost_gain_groups=5
no_cost_gain_explain_rows=5
violates_feasible_constraints=0
no_action_delta=0
accepted_candidate=36
selected=23
unselected=33
hard_gate_reasons=detailing_ratio_above_hard_limit=3

Selected Cost-Down Candidates

S04 intermediate wall rebar_down | gain=154.375 | trial_dcr=0.986 | selected_constructability_gain_after_cost_pass
S03 perimeter wall rebar_down | gain=154.375 | trial_dcr=0.986 | selected_constructability_gain_after_cost_pass
S05 perimeter column perimeter_frame_down | gain=47.994 | trial_dcr=-0.613 | selected_constructability_gain_after_cost_pass
S04 perimeter slab slab_thickness_down | gain=7.402 | trial_dcr=0.000 | selected_constructability_gain_after_cost_pass
S02 perimeter wall wall_thickness_down | gain=4.470 | trial_dcr=0.986 | selected_constructability_gain_after_cost_pass
S04 intermediate wall wall_thickness_down | gain=4.346 | trial_dcr=0.986 | selected_constructability_gain_after_cost_pass
S07 intermediate wall wall_thickness_down | gain=4.346 | trial_dcr=0.986 | selected_constructability_gain_after_cost_pass
S03 intermediate wall wall_thickness_down | gain=4.346 | trial_dcr=0.986 | selected_constructability_gain_after_cost_pass
S06 intermediate wall wall_thickness_down | gain=4.346 | trial_dcr=0.986 | selected_constructability_gain_after_cost_pass
S04 perimeter wall detailing_down | gain=3.199 | trial_dcr=0.986 | selected_constructability_gain_after_cost_pass
S03 perimeter wall detailing_down | gain=3.199 | trial_dcr=0.986 | selected_constructability_gain_after_cost_pass

S04 perimeter wall detailing_down | gain=3.199 | trial_dcr=0.986 | selected_constructability_gain_after_cost_pass
S07 perimeter wall detailing_down | gain=3.199 | trial_dcr=0.986 | selected_constructability_gain_after_cost_pass
S06 perimeter wall detailing_down | gain=3.199 | trial_dcr=0.986 | selected_constructability_gain_after_cost_pass
S08 perimeter wall detailing_down | gain=2.940 | trial_dcr=0.000 | selected_constructability_gain_after_cost_pass
S04 transfer beam connection_detailing_down | gain=0.355 | trial_dcr=0.986 | selected_constructability_gain_after_cost_pass
S04 transfer beam connection_detailing_down | gain=0.355 | trial_dcr=0.986 | selected_constructability_gain_after_cost_pass
S04 transfer beam connection_detailing_down | gain=0.355 | trial_dcr=0.986 | selected_constructability_gain_after_cost_pass
S04 transfer beam connection_detailing_down | gain=0.355 | trial_dcr=0.986 | selected_constructability_gain_after_cost_pass
S04 transfer beam connection_detailing_down | gain=0.355 | trial_dcr=0.986 | selected_constructability_gain_after_cost_pass
S04 perimeter slab slab_thickness_down | gain=7.468 | trial_dcr=0.986 | selected_constructability_gain_after_cost_pass
S02 perimeter wall wall_thickness_down | gain=4.470 | trial_dcr=0.986 | selected_constructability_gain_after_cost_pass
S02 perimeter wall wall_thickness_down | gain=4.470 | trial_dcr=0.986 | selected_constructability_gain_after_cost_pass

Unselected Cost-Down Candidates

S08 perimeter wall rebar_down | gain=154.375 | trial_dcr=0.000 | rejected_batch_budget_limit
S09 perimeter wall rebar_down | gain=154.375 | trial_dcr=0.000 | rejected_batch_budget_limit
S09 perimeter wall rebar_down | gain=154.375 | trial_dcr=0.000 | rejected_batch_budget_limit
S08 perimeter wall rebar_down | gain=154.375 | trial_dcr=0.000 | rejected_batch_budget_limit
S04 perimeter slab rebar_down | gain=102.917 | trial_dcr=0.000 | rejected_batch_budget_limit
S09 perimeter slab rebar_down | gain=102.917 | trial_dcr=0.000 | rejected_batch_budget_limit
S04 intermediate slab rebar_down | gain=101.588 | trial_dcr=0.000 | rejected_batch_budget_limit
S09 intermediate slab rebar_down | gain=101.588 | trial_dcr=0.000 | rejected_batch_budget_limit
S04 core slab rebar_down | gain=100.791 | trial_dcr=0.000 | rejected_batch_budget_limit
S09 core slab rebar_down | gain=100.791 | trial_dcr=0.000 | rejected_batch_budget_limit
S05 perimeter column rebar_down | gain=47.994 | trial_dcr=-0.613 | rejected_batch_budget_limit
S05 intermediate column rebar_down | gain=47.994 | trial_dcr=-0.553 | rejected_batch_budget_limit
S01 intermediate foundation rebar_down | gain=31.266 | trial_dcr=0.000 | rejected_batch_budget_limit
S00 core foundation rebar_down | gain=31.266 | trial_dcr=0.000 | rejected_batch_budget_limit

S09 perimeter slab slab_thickness_down | gain=7.402 | trial_dcr=0.000 | rejected_batch_budget_limit
S04 intermediate slab slab_thickness_down | gain=7.307 | trial_dcr=0.000 | rejected_batch_budget_limit
S09 intermediate slab slab_thickness_down | gain=7.307 | trial_dcr=0.000 | rejected_batch_budget_limit
S04 core slab slab_thickness_down | gain=7.249 | trial_dcr=0.000 | rejected_batch_budget_limit
S09 core slab slab_thickness_down | gain=7.249 | trial_dcr=0.000 | rejected_batch_budget_limit
S01 intermediate foundation detailing_down | gain=4.618 | trial_dcr=0.000 | rejected_batch_budget_limit
S08 perimeter wall wall_thickness_down | gain=4.339 | trial_dcr=0.000 | rejected_batch_budget_limit
S09 perimeter wall wall_thickness_down | gain=4.339 | trial_dcr=0.000 | rejected_batch_budget_limit
S09 perimeter wall wall_thickness_down | gain=4.339 | trial_dcr=0.000 | rejected_batch_budget_limit
S08 perimeter wall wall_thickness_down | gain=4.339 | trial_dcr=0.000 | rejected_batch_budget_limit
S04 perimeter slab detailing_down | gain=2.940 | trial_dcr=0.000 | rejected_batch_budget_limit
S09 perimeter slab detailing_down | gain=2.940 | trial_dcr=0.000 | rejected_batch_budget_limit
S08 perimeter wall detailing_down | gain=2.940 | trial_dcr=0.000 | rejected_batch_budget_limit
S09 perimeter wall detailing_down | gain=2.940 | trial_dcr=0.000 | rejected_batch_budget_limit
S09 perimeter wall detailing_down | gain=2.940 | trial_dcr=0.000 | rejected_batch_budget_limit
S04 intermediate slab detailing_down | gain=2.903 | trial_dcr=0.000 | rejected_batch_budget_limit
S09 intermediate slab detailing_down | gain=2.903 | trial_dcr=0.000 | rejected_batch_budget_limit
S04 core slab detailing_down | gain=2.880 | trial_dcr=0.000 | rejected_batch_budget_limit
S09 core slab detailing_down | gain=2.880 | trial_dcr=0.000 | rejected_batch_budget_limit
S00 core foundation detailing_down | gain=0.000 | trial_dcr=0.000 | rejected_lower_cost_gain
S04 transfer beam connection_detailing_down | gain=0.000 | trial_dcr=0.000 | rejected_lower_cost_gain
S04 transfer beam beam_section_down | gain=4.618 | trial_dcr=0.933 | rejected_detailing_hard_gate
S04 transfer beam rebar_down | gain=45.696 | trial_dcr=0.933 | rejected_detailing_hard_gate
S04 transfer beam detailing_down | gain=0.000 | trial_dcr=0.000 | rejected_lower_cost_gain
S05 perimeter column connection_detailing_down | gain=0.000 | trial_dcr=-0.613 | rejected_lower_cost_gain
S05 perimeter column detailing_down | gain=0.000 | trial_dcr=-0.613 | rejected_lower_cost_gain
S05 intermediate column detailing_down | gain=0.000 | trial_dcr=-0.553 | rejected_lower_cost_gain
S05 core column connection_detailing_down | gain=0.000 | trial_dcr=-0.441 | rejected_lower_cost_gain

S05 core column rebar_down | gain=47.994 | trial_dcr=0.987 | rejected_detailing_hard_gate

S05 core column detailing_down | gain=0.000 | trial_dcr=-0.441 | rejected_lower_cost_gain

Design Change Summary

S02 total | changed=1 semantic=0 | cost=-13.203 | dcr=0.000->0.976

S03 total | changed=3 semantic=0 | cost=-161.655 | dcr=0.000->0.550

S04 total | changed=10 semantic=1 | cost=-170.808 | dcr=0.498->0.507

S05 total | changed=1 semantic=1 | cost=-47.994 | dcr=0.000->0.239

S06 total | changed=2 semantic=0 | cost=-7.280 | dcr=0.000->0.550

S07 total | changed=2 semantic=0 | cost=-7.280 | dcr=0.000->0.550

S08 total | changed=1 semantic=0 | cost=-2.940 | dcr=0.000->0.504

intermediate total | changed=5 semantic=0 | cost=-171.732 | dcr=0.000->0.487

perimeter total | changed=10 semantic=2 | cost=-248.085 | dcr=0.000->0.976

transfer total | changed=5 semantic=0 | cost=8.658 | dcr=0.498->0.352

S02 perimeter wall | changed=1 semantic=0 | cost=-13.203 | dcr=0.000->0.976 | constructability=0.4232->0.3212 | gate=hard_gate_pass

S03 intermediate wall | changed=1 semantic=0 | cost=-4.339 | dcr=0.000->0.487 | constructability=0.4106->0.3766 | gate=hard_gate_pass

S03 perimeter wall | changed=2 semantic=0 | cost=-157.316 | dcr=0.000->0.550 | constructability=0.4194->0.3734 | gate=hard_gate_pass

S04 intermediate wall | changed=2 semantic=0 | cost=-158.714 | dcr=0.000->0.412 | constructability=0.4106->0.3836 | gate=hard_gate_pass

S04 perimeter slab | changed=1 semantic=1 | cost=-14.871 | dcr=0.000->0.419 | constructability=0.1084->0.0684 | gate=hard_gate_pass

S04 perimeter wall | changed=2 semantic=0 | cost=-5.881 | dcr=0.000->0.507 | constructability=0.4194->0.3474 | gate=hard_gate_pass

S04 transfer beam | changed=5 semantic=0 | cost=8.658 | dcr=0.498->0.352 | constructability=0.4675->0.3955 | gate=hard_gate_pass

S05 perimeter column | changed=1 semantic=1 | cost=-47.994 | dcr=0.000->0.239 | constructability=0.4779->0.4499 | gate=hard_gate_pass

S06 intermediate wall | changed=1 semantic=0 | cost=-4.339 | dcr=0.000->0.455 | constructability=0.4106->0.3766 | gate=hard_gate_pass

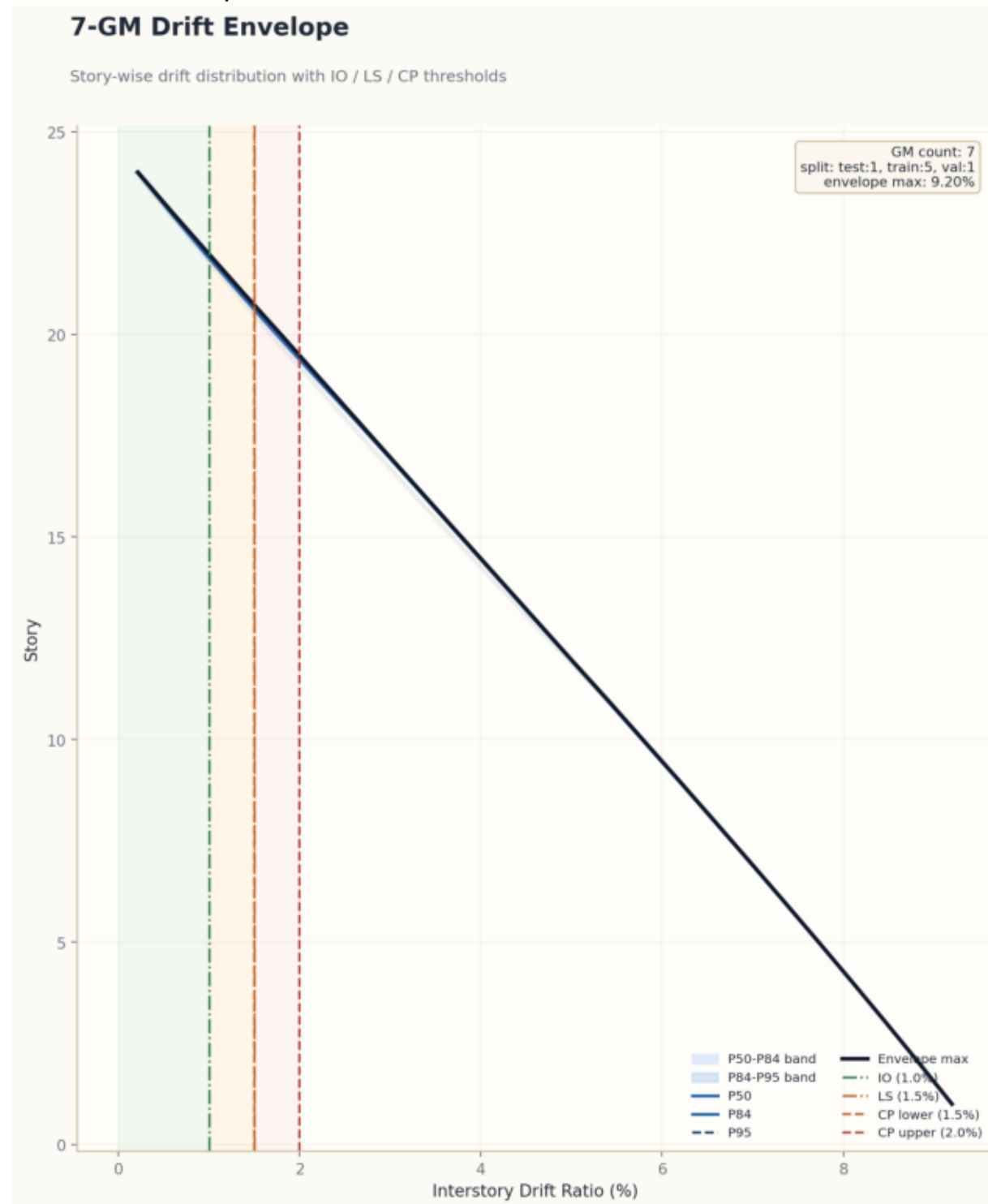
S06 perimeter wall | changed=1 semantic=0 | cost=-2.940 | dcr=0.000->0.550 | constructability=0.4194->0.3474 | gate=hard_gate_pass

S07 intermediate wall | changed=1 semantic=0 | cost=-4.339 | dcr=0.000->0.455 | constructability=0.4106->0.3766 | gate=hard_gate_pass

S07 perimeter wall | changed=1 semantic=0 | cost=-2.940 | dcr=0.000->0.550 | constructability=0.4194->0.3474 | gate=hard_gate_pass

S08 perimeter wall | changed=1 semantic=0 | cost=-2.940 | dcr=0.000->0.504 | constructability=0.4147->0.3427 | gate=hard_gate_pass

Drift Envelope



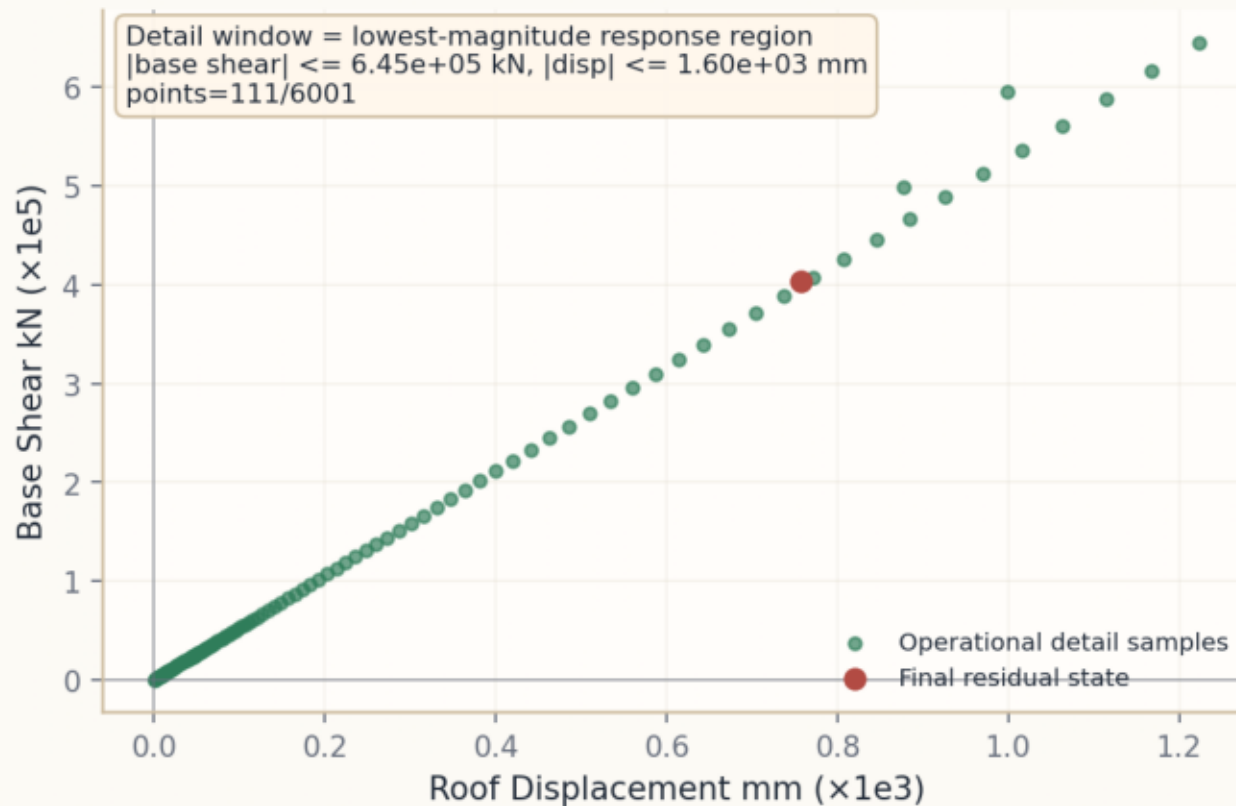
Core Hysteresis

System Hysteresis

Representative core loop and residual-state evidence | opstool_606m_megatall_model-00001

Overflow / unstable regime

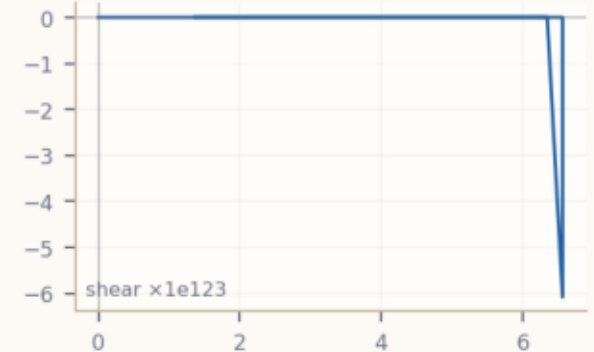
Solver instability suspicion



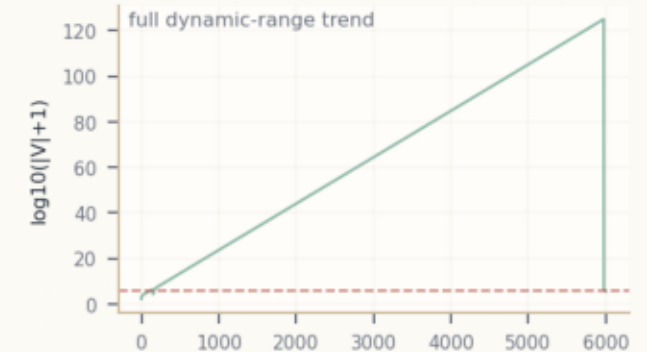
Residual drift: 1.914%
Residual top disp: 757.74 mm
Cumulative E_{diss} : $6.369\text{e}+126$ kN·mm

Overflow regime detected.
Main panel shows only the lowest-magnitude operational window.
Read the right-hand magnitude panel for the full dynamic-range trend.

Core loop detail

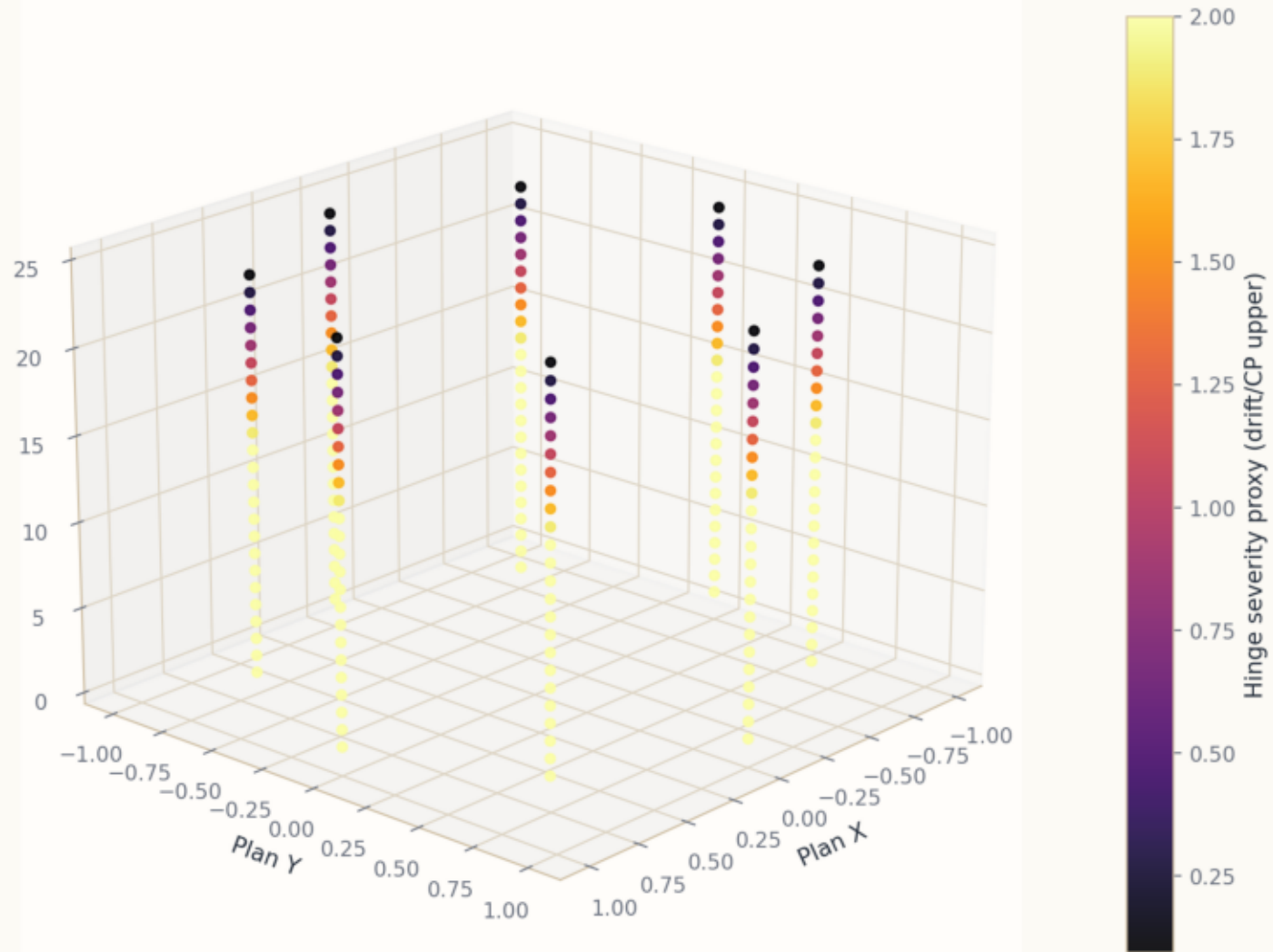


Magnitude by step

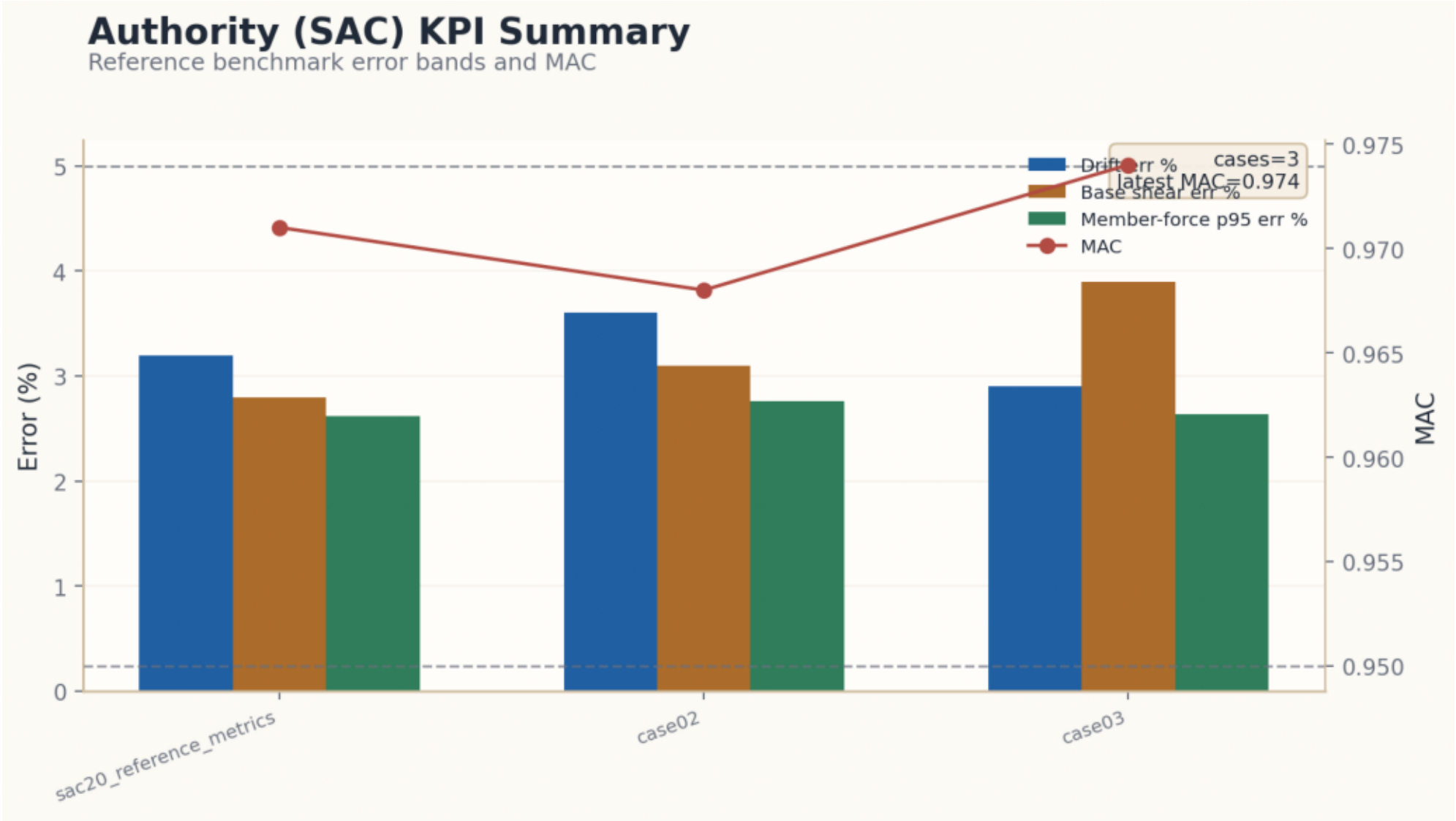


Plastic Hinge Proxy

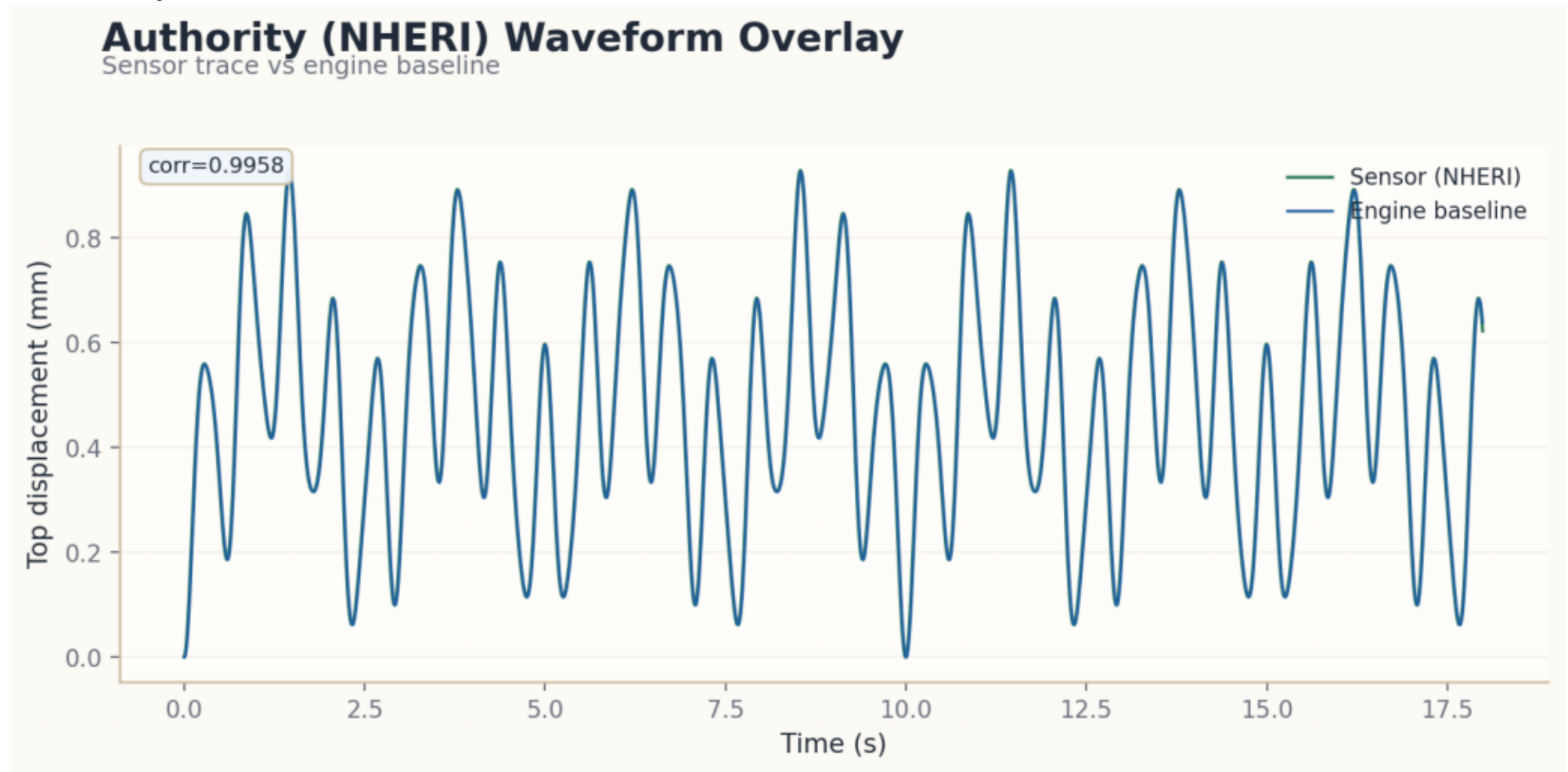
3D Plastic-Hinge Proxy (opstool_606m_megatall_model-00001)



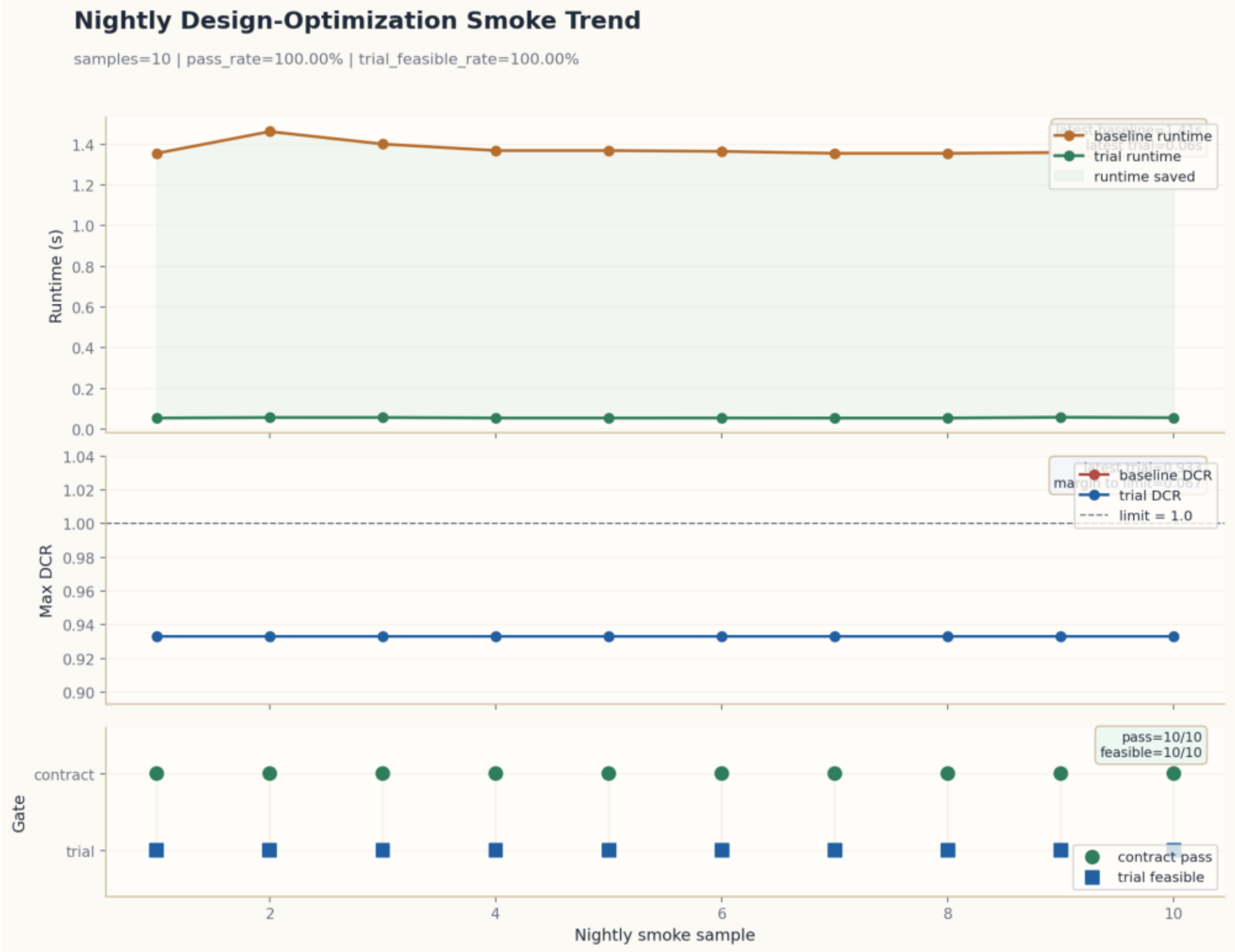
Authority SAC KPI



Authority NHERI Waveform



Nightly Smoke Trend



Nightly Smoke Recent Samples

#6 2026-04-06T12:19:39.039411+00:00 | pass=True | trial_feasible=True | baseline_runtime=1.3646s | trial_runtime=0.0554s | trial_max_dcr=0.9332 | action=connect

#7 2026-04-06T12:23:56.108705+00:00 | pass=True | trial_feasible=True | baseline_runtime=1.3550s | trial_runtime=0.0550s | trial_max_dcr=0.9332 | action=connect

#8 2026-04-06T12:23:56.108705+00:00 | pass=True | trial_feasible=True | baseline_runtime=1.3550s | trial_runtime=0.0550s | trial_max_dcr=0.9332 | action=connect

#9 2026-04-06T12:32:45.009996+00:00 | pass=True | trial_feasible=True | baseline_runtime=1.3590s | trial_runtime=0.0590s | trial_max_dcr=0.9332 | action=connect

#10 2026-04-06T13:54:47.180033+00:00 | pass=True | trial_feasible=True | baseline_runtime=1.4059s | trial_runtime=0.0567s | trial_max_dcr=0.9332 | action=connect

Residual Holdout Boundary

Licensed Engineer Review | owner=기술사 | share=50% | project=0.5-2.5% | non-standard interpretation, final judgment, exceptional irregularity, and member-level edge cases

Legacy Tool Cross-Validation | owner=기존툴+기술사 | share=30% | project=0.3-1.5% | novel load paths, authority-critical submodels, and residual niche workflows outside standard toolsets

Legal Sign-Off | owner=기술사/기존 승인 workflow | share=20% | project=0.2-1.0% | formal seal, legal submission, and authority-facing responsibility that remains outside technical review

Time-Saving Coverage

Coverage target: 95-99%

Residual holdout: 1-5%

Estimated time saved: 90-96%

Measured chain wall-clock (comparable rolling): 0.88 min

Rolling sample count: 14, range=0.846-0.906 min

Measured chain wall-clock (current): 0.88 min

Comparable run mode: `current_pipeline_comparable_full_chain_pass`

Empirical smoke runtime reduction: 95.66-96.04%

Basis: Empirical estimate derived from nightly design-optimization smoke runtime reduction, scaled by the accelerated-coverage target.
`smoke_mean_runtime_saved=95.92%`, `sample_count=10`.

Use this engine to automate the dominant, time-consuming 95-99% of repeated analysis, screening, packaging, and optimization workflows. Keep the residual 1-5% under licensed engineer review, legacy-tool cross-check, and formal sign-off workflows.

Advanced Holdout Closure

total=4 | closed=4 | open=0 | closed=4/4 | open=0 | severities=P0:2, P1:2

Dynamic plastic-hinge refresh | severity=P0 | closure=closed | mode=computed_member_local_hinge_refresh

reason=Dynamic hinge-refresh artifact is attached. | evidence=hinge_proxy_artifacts=2, artifact_present=True, artifact_kind=hinge_refresh_projected_from_optimization_changes, source_mode=rebar_sensitive_member_local_refresh, overlap_members=88, rebar_sensitive_members=70, benchmark_assets=5, benchmark_split=train:2/va...

Panel-zone 3D clash and anchorage coverage | severity=P0 | closure=closed | mode=panel_zone_3d_clash_and_anchorage_verified

reason=3D panel-zone clash and anchorage recomputation artifacts are attached | evidence=proxy_candidates=45, source=design_optimization_dataset_npz:true_3d_clash_and_anchorage_verified, status_label=release_ready, advisory_only=False, release_blocking=False, artifact_closed=True, closure_mode=closed_exact_validated, coverage=validated_sources=...

Foundation / mat / pile optimization | severity=P1 | closure=closed | mode=active_foundation_member_optimization

reason=foundation optimization artifact is attached and dataset contains foundation members | evidence=foundation_member_type_count=76, scope_source=dataset_summary, raw_source_labels=0, upstream_labels=0, upstream_mode=dataset_scope_only

Raw wind-tunnel data mapping | severity=P1 | closure=closed | mode=raw_hffb_node_pressure_mapping

reason=Raw wind-tunnel HFFB mapping is ready for traceable MIDAS binding. | evidence=semantic_pressure_binding=True, bound_pressure_rows=7278, unbound_pressure_rows=0

Residual Holdout Review Table

Licensed Engineer Review | review_story_zone | S04/perimeter (2), S02/perimeter (1), S03/intermediate (1), S03/perimeter (1)

owner=기술사 | why=Top story-zone review pockets are derived from actual accepted design-change rows so engineer holdout stays tied to the highest-touch parts of the structure.

Licensed Engineer Review | story_band | 4, 3, 6

owner=기술사 | why=High-touch story bands remain under engineer review because they concentrate accepted design changes and irregular response checks.

Licensed Engineer Review | member_family | wall (26), slab (19), column (9)

owner=기술사 | why=Dominant member families in accepted optimization changes still require engineer judgment on local edge cases and detailing intent.

Licensed Engineer Review | zone | perimeter (8), intermediate (4), transfer (1)

owner=기술사 | why=Zone concentration is used to focus manual review on the highest-touch portions of the structural layout.

Legacy Tool Cross-Validation | submodel_family | SCBF16B (3), SCBF16B_shell_beam_mix (2), opensees (1), nheri_case01_sensor (1)

owner=기존툴+기술사 | why=Authority submodel families are derived from the active catalog paths so cross-validation follows the exact benchmark submodels still outside the accelerated envelope.

Legacy Tool Cross-Validation | authority_critical_case | OpenSees (3), SAC (3), NHERI (3)

owner=기존툴+기술사 | why=Authority-critical benchmark tracks remain the primary cross-validation target outside the accelerated envelope.

Legacy Tool Cross-Validation | authority_catalog_case_id | SAC20_LA_holdout_01, SAC20_SEA_holdout_02, SAC20_BOS_holdout_03, NHERI_UCSD_like_holdout

owner=기존툴+기술사 | why=Authority catalog case ids are read directly so the holdout review list refreshes automatically when the benchmark catalog changes.

Legal Sign-Off | authority_catalog_track | opensees (3), sac (3), nheri (3)

owner=기술사/기존 승인 workflow | why=Formal authority-facing responsibility is anchored to the active authority catalog tracks and remains outside the automated responsibility boundary.

Legal Sign-Off | authority_critical_case | sealed submission pack, authority-facing variants, stamped final issue

owner=기술사/기존 승인 workflow | why=Formal authority-facing deliverables remain outside the automated responsibility boundary.

Residual Holdout Routing Matrix

opensees / SCBF16B -> S04/perimeter / wall

owner=기존툴+기술사 | why=Routing matrix links active authority submodels to dominant story/zone/member-family review pockets so the remaining manual and legacy-tool work stays explicit.

opensees / SCBF16B_shell_beam_mix -> S02/perimeter / slab

owner=기존툴+기술사 | why=Routing matrix links active authority submodels to dominant story/zone/member-family review pockets so the remaining manual and legacy-tool work stays explicit.

opensees / opensees -> S03/intermediate / column

owner=기존툴+기술사 | why=Routing matrix links active authority submodels to dominant story/zone/member-family review pockets so the remaining manual and legacy-tool work stays explicit.

sac / SCBF16B -> S03/perimeter / beam

owner=기존툴+기술사 | why=Routing matrix links active authority submodels to dominant story/zone/member-family review pockets so the remaining manual and legacy-tool work stays explicit.

sac / SCBF16B_shell_beam_mix -> S04/perimeter / wall

owner=기존툴+기술사 | why=Routing matrix links active authority submodels to dominant story/zone/member-family review pockets so the remaining manual and legacy-tool work stays explicit.

nheri / nheri_case01_sensor -> S02/perimeter / slab

owner=기존툴+기술사 | why=Routing matrix links active authority submodels to dominant story/zone/member-family review pockets so the remaining manual and legacy-tool work stays explicit.

Authority Catalog Routing Diff

baseline_seeded=False | changes=0 | added=0 | removed=0 | unchanged=8

No authority-catalog routing changes detected for this package refresh.